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Understanding MLP-Mixer as a Wide and Sparse MLP

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Abstract

Multi-layer perceptron (MLP) is a fundamental component of deep learning, and recent MLP-based architectures, especially the MLP-Mixer, have achieved significant empirical success. Nevertheless, our understanding of why and how the MLP-Mixer outperforms conventional MLPs remains largely unexplored. In this work, we reveal that sparseness is a key mechanism underlying the MLP-Mixers. First, the Mixers have an effective expression as a wider MLP with Kronecker-product weights, clarifying that the Mixers efficiently embody several sparseness properties explored in deep learning. In the case of linear layers, the effective expression elucidates an implicit sparse regularization caused by the model architecture and a hidden relation to Monarch matrices, which is also known as another form of sparse parameterization. Next, for general cases, we empirically demonstrate quantitative similarities between the Mixer and the unstructured sparse-weight MLPs. Following a guiding principle proposed by Golubeva, Neyshabur and Gur-Ari (2021), which fixes the number of connections and increases the width and sparsity, the Mixers can demonstrate improved performance.

1. Introduction

Multi-layer perceptron (MLP) and its variants are fundamental components of deep learning employed in various problems and for understanding the basic properties of neural networks. Despite their simplicity and long history (Rosenblatt, 1958; Schmidhuber, 2015), it has become apparent only recently that there is still significant room for improvement in the predictive performance of MLP-based architectures.

The sparseness is known as a key direction for enhancing the performance of dense MLP layers (Neyshabur et al., 2014; d'Ascoli et al., 2019; Neyshabur, 2020; Golubeva et al., 2021; Pellegrini & Biroli, 2022). For instance, Golubeva et al. (2021) reported that the prediction performance improves by increasing both the width and the sparsity of connectivity when the number of trainable parameters is fixed. The MLP-Mixer is another noteworthy direction of recent developments in MLP-based architectures (Tolstikhin et al., 2021; Touvron et al., 2022). It does not rely on convolutions or self-attention and is entirely composed of MLP layers; instead, it utilizes MLPs applied across spatial locations or feature channels. This can be regarded as an MLP that applies fully connected (FC) layers on both sides of the feature matrix. Despite its simplicity, the MLP-Mixer achieved a performance on image classification benchmarks comparable to that of more structured deep neural networks.

However, there are few studies on experimental and theoretical attempts to understand MLP-Mixer's internal mechanisms (Yu et al., 2022; Sahiner et al., 2022). This contrasts with the extensive elucidation of explicit or implicit biases in other modern architectural components (Neyshabur et al., 2014; Bjorck et al., 2018; Cordonnier et al., 2019). To further advance the MLP-based architecture, it will be crucial to unveil the underlying mechanism of the MLP-Mixer and to address questions such as: What different inductive biases does the MLP-Mixer have compared to a naive MLP? Which architectural factors significantly contribute to their superior performance?

In this study, we reveal that the sparseness, which is seemingly a distinct research concept, is the key mechanism underlying the MLP-Mixer. One can see that the MLP-Mixer is a wider MLP with sparsity inherently embedded as an inductive bias. This equivalence also provides a quantitative understanding that appropriate token and channel sizes maximizing the sparsity can improve the prediction performance. The detailed contributions are summarized as follows:

- We first identify *an effective expression of MLP-Mixer as an MLP* by vectorizing the mixing layers (in Section 3). It is composed of the permutation matrix and the Kronecker product and provides an interpretation of mixing layers as an extremely wide MLP with sparse (structured) weights.

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- We consider a linear activation case of a simple Mixer and its effective expression. First, this reveals that the Kronecker-product weight possesses a bias towards an implicit L1 regularization (in Section 3.2). In deep learning, there are two aspects of sparsity: one is a large number of zero parameters (e.g., Neyshabur (2020)) and the other is the limited number of independent parameters (e.g., Dao et al. (2022)). Our evaluation of implicit regularization effectively links these two facets of sparsity. Second, regarding the limited number of independent parameters, the MLP-Mixer can be regarded as an approximation of an MLP with the Monarch matrix (in Section 3.3). Thus, the sparseness discussed in the different literature is inherently and originally incorporated into the Mixers.

- For realistic cases with non-linear activation functions, we quantitatively evaluate the high similarity between the traditional sparse-weight MLP (SW-MLP) and the MLP-Mixer. First, we confirm the similarity of hidden features by the centered kernel alignment. Second, we reveal similar performance trends when increasing sparseness (equivalent to widening) with a fixed number of connections. This means that empirical observations of Golubeva et al. (2021) on appropriate sizes of sparsity hold even in the MLP-Mixer in the sense of both improving prediction performance (in Section 4.2) and ensuring trainability (in Section 4.3). We also empirically verify that the Random-Permuted (RP) Mixer introduced in Section 5.2, which is a less structured variant of the normal Mixer, exhibits similar performance trends (Sections 5.3 & 5.4). This further solidifies our understanding that sparsity is a fundamental element underlying the Mixers.

2. Preliminaries

2.1. Related Work

MLP-based architectures. The salient property of an MLP-Mixer is that it is composed entirely of FC layers. This property is unique to the MLP-Mixer (and its concurrent work ResMLP (Touvron et al., 2022)) and different from attention-based architectures (Dosovitskiy et al., 2021). While some previous work focused on providing a relative evaluation of performance compared with the attention module (Yu et al., 2022; Sahiner et al., 2022), our purpose is to elucidate the hidden bias of MLP-Mixers as a wide and sparse MLP. Golubeva et al. (2021) investigated that the generalization performance can be improved by increasing the width in FC layers. Because they fixed the number of weights, an increase in the width caused a higher sparsity. They revealed that even for fixed sparse connectivity throughout training, a large width can improve the performance better than the dense layer.

Structured weight matrices: (i) Sparse matrix. Parameter sparsity is widely used to improve the performance and efficiency of neural networks. It is known that naive MLPs require even more data than structured deep models to improve performance due to their weak inductive bias (Bachmann et al., 2023). The sparseness is considered useful as a minimal necessary bias. One approach to make weights sparse is to determine nonzero weights dynamically, such as dense-to-sparse training (Neyshabur, 2020), pruning (Frankle & Carbin, 2019), and sparse-to-sparse training (Dettmers & Zettlemoyer, 2019; Evci et al., 2020). The other is to constrain the trainable weights from the beginning of training statically (Dao et al., 2022; Golubeva et al., 2021; Liu et al., 2022; Gadhikar et al., 2023). The current study follows the latter approach; specifically, we reveal that the mixing layers of the MLP-Mixer are implicitly related to such fixed sparse connectivity. **(ii) Kronecker product.** Constraining weight matrices to the Kronecker product and its summation has been investigated in the model-compression literature. Some works succeeded in reducing the number of trainable parameters without deteriorating the prediction performance (Zhou et al., 2015; Zhang et al., 2021) while others applied them for the compression of trained parameters (Hameed et al., 2022). In contrast, we find a Kronecker product expression hidden in the MLP-Mixer, which can be regarded as an approximation of the Monarch matrix proposed in Dao et al. (2022).

Notably, our study is completely different from merely applying sparse regularizers, Kronecker-product weights, or Monarch matrices (Dao et al., 2022) to the dense weight matrices in the mixing layers like Fu et al. (2023). Our finding is that MLP-Mixers and their generalization (i.e., the PK family) inherently possess these properties.

2.2. Notations

MLP-Mixer. An MLP-Mixer is defined as follows (Tolstikhin et al., 2021). Initially, it divides an input image into patches. Next, a per-patch FC layer is performed. After that, the blocks described as follows are repeatedly applied to them: for the feature matrix from the previous hidden layer $X \in \mathbb{R}^{S \times C}$,

$$\text{Token-MLP}(X) = W_2 \phi(W_1 X), \tag{1}$$

$$\text{Channel-MLP}(X) = \phi(X W_3) W_4, \tag{2}$$

where ϕ denotes the entry-wise activation function, $W_1 \in \mathbb{R}^{\gamma S \times S}$, $W_2 \in \mathbb{R}^{S \times \gamma S}$, $W_3 \in \mathbb{R}^{C \times \gamma C}$ and $W_4 \in \mathbb{R}^{\gamma C \times C}$. In this paper, we set the expansion factor of the hidden layers of token and channel-mixing MLPs to the same value γ for simplicity. The block of the MLP-Mixer is given by the map $X \mapsto Y$, where

$$U = X + \text{Token-MLP}(\text{LN}(X)), \tag{3}$$

$$Y = U + \text{Channel-MLP}(\text{LN}(U)). \tag{4}$$

In the end, the global average pooling and the linear classifier are applied to the last hidden layer.

S-Mixer. To facilitate theoretical insights, we introduce the S-Mixer, an idealized version of the MLP Mixer:

$$H = \phi(\phi(WX)V), \tag{5}$$

where V and W represent weight matrices. In this formulation, the shallow neural networks typically found within the mixing block of the MLP-Mixer are simplified to single layers. For the sake of simplicity in our theoretical analysis of the S-Mixer, we omit layer normalization and skip connections. It is important to note, however, that in our numerical experiments on training deep models in Section 5, these components are incorporated, demonstrating that their inclusion does not detract from the fundamental outcomes of the study. we summarized the detailed architectures used for experiments in Appendix A.

3. Properties as a sparse MLP

3.1. Vectorization

To address the similarity between MLP-Mixer and MLP, we consider vectorization of feature tensors and effective width. We represent the vectorization operation of the matrix $X \in \mathbb{R}^{S\times C}$ by $\mathrm{vec}(X)$; more precisely, $(\mathrm{vec}(X))_{(j-1)d+i} = X_{ij}, (i = 1, \ldots, S, j = 1, \ldots, C)$. We also define an inverse operation $\mathrm{mat}(\cdot)$ to recover the matrix representation by $\mathrm{mat}(\mathrm{vec}(X)) = X$. There exists a well-known equation for the vectorization operation and the Kronecker product denoted by $\otimes$;

$$\mathrm{vec}(WXV) = (V^\top \otimes W)\mathrm{vec}(X), \tag{6}$$

for $W \in \mathbb{R}^{S\times S}$ and $V \in \mathbb{R}^{C\times C}$. The vectorization of the feature matrix WXV is equivalent to a fully connected layer of width $m := SC$ with a weight matrix $V^\top \otimes W$. We refer to this m as the *effective width* of mixing layers.

In MLP-Mixer, when we treat each $S \times C$ feature matrix X as an SC-dimensional vector $\mathrm{vec}(X)$, the right multiplication by an $C\times C$ weight V and the left weight multiplication by a $S \times S$ weight W are represented as follows:

$$\mathrm{vec}(XV) = (V^\top \otimes I_S)\mathrm{vec}(X), \tag{7}$$

$$\mathrm{vec}(WX) = (I_C \otimes W)\mathrm{vec}(X) \tag{8}$$

where I_n denotes an $n \times n$ identity matrix. This expression clarifies that the mixing layers work as an MLP with special weight matrices with the Kronecker product. As usual, the size of S and C is approximately $10^2 \sim 10^3$, and this implies that the Mixer is equivalent to an extremely wide MLP with $m = 10^4 \sim 10^6$. Moreover, the ratio of non-zero entries in the weight matrix $I_C \otimes W$ is $1/C$ and that of $V^\top \otimes I_S$ is $1/S$. Therefore, the weight of the effective MLP is highly sparse in the sense of non-zero entries.

Here, to consider only the left-multiplication of weights, we introduce commutation matrices.

Definition 3.1. The commutation matrix J_c is an $m \times m$ matrix defined as

$$J_c\mathrm{vec}(X) = \mathrm{vec}(X^\top), \tag{9}$$

where X is an $S \times C$ matrix.

Note that for any $x \in \mathbb{R}^m$, $J_c\phi(x) = \phi(J_c x)$. In addition, we have $J_c^\top(I_S \otimes V^\top)J_c = V^\top \otimes I_S$ for any $C \times C$ matrix V. Using the commutation matrix, we find the following:

Proposition 3.2 (Effective expression of MLP-Mixer as MLP)**.** *The feature matrix of the S-Mixer (5) is a shallow MLP with width $m = SC$ as follows:*

$$vec(H) = \phi\left(J_c^\top\left(I_S \otimes V^\top\right)\phi\left(J_c\left(I_C \otimes W\right)vec\left(X\right)\right)\right). \tag{10}$$

The derivation is straightforward as described in Appendix B.1. This expression clarifies that the mixing layers work as an MLP with special weight matrices with the commutation matrix and Kronecker product.

It is easy to generalize the above expression for the S-Mixer to the MLP-Mixer, where each mixing operation is composed of shallow neural networks (2) (see Appendix B.1). This equivalence with a wide MLP with sparse weights is simple and easy to follow but has been missing in the literature.

3.2. Implicit regularization of linear mixing layers

Considering the linear activation case to gain theoretical insight is a common approach used in deep learning (Saxe et al., 2014; Arora et al., 2019). yet there has been no work on the Mixers. We find that the theoretical analysis is challenging even in the simplest case of the linear S-Mixer (6). This model is equivalent to a so-called bi-linear regression model, for which an analytical solution that minimizes the MSE loss is unknown (Hoff, 2015). This makes analyzing the linear S-Mixer more difficult compared to conventional linear networks. Despite this difficulty, we discover the following inequality that characterizes the implicit regularization of the model:

Proposition 3.3.

$$\min_{V,W} \mathcal{L}(V \otimes W) + \frac{\lambda}{2}(\|V\|_F^2 + \|W\|_F^2) \geq \min_{B\in\mathbb{R}^{SC\times SC}} \mathcal{L}(B) + \tilde{\lambda}\|B\|_1 \text{ with } \tilde{\lambda} = \lambda/CS \tag{11}$$

where $\|\cdot\|_F$ is the Frobenius norm, $\|\cdot\|_1$ is the L1 norm.